Canyon Run (Canyon / Highway / Mobility, Ambushes, Route Control)

Executive Summary

Canyon Run is a **hallucinated canyon-highway battlefield** built for **mobility-focused combat. ambush tactics. route control. and autonomous maneuvering through narrow vertical terrain**. The environment features winding canyon highways. cliffside overhanes. collapsed tunnels. dry riverbeds. abandoned checkpoints. and elevated ridelines. The mas-emphasizes Obsidian Protocol's autonomous route-selection laalc. sensor-fusion under dust and heat distortion. comms-relay warfare. and logistics under fast-movine highway conditions. In AR (Nreal Light)/Art. the mas anchors at ~4–5 m viewine distance. coverine a 10×10 m shvsical area nepresentine ~1 km² of canvon-highway terrain. Kev features include ambush corridors. high-speed traversal lanes. cliffside overwatch. and choke-controlled mobility warfare.

Design Goals & Gameplay

Mobility-Driven Combat
Canyon Run prioritizes fast movement, convoy operations, and rapid flanking. Highway lanes allow high-speed traversal; canyon walls restrict lateral movement.

Ambush & Route Control
Narrow passes, blind corners, and cliff overhangs create ideal ambush zones. Controlling key segments of the highway determines reinforcement flow.

Vertical Overwatch & Canyon Tactics
Ridges and cliffs provide overwatch positions for snipers, artillery, and drones. Verticality shapes sightlines and engagement ranges.

Autonomy & Route-Selection Logic
Autonomous units choose between highway lanes, cliffside trails, dry riverbeds, and tunnel bypasses. Units adapt to blocked roads, collapsed tunnels, and shifting ambush threats.

Sensor & Information Warfare
Dust storms, heat shimmer, canyon occlusion, and thermal plumes degrade sensors. Relay towers on ridges restore clarity. Destroying relays creates blind pockets across the canyon.

Logistics & Resources
Fuel depots, checkpoints, and roadside maintenance yards serve as resupply nodes. Convoys must navigate exposed roads and avoid ambush-prone canyon passes.

Deployment & Progress
A fixed deployment budget (~10,000 points) forces players to choose between fast armor, ambush specialists, or long-range overwatch units. Blue deploys from the southwest canyon highway gate; Red deploys from the northeast ridge-top command.

Spatial Constraints (Nreal AR)

Device Capabilities
Nreal Light/Air; ~52° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling
10×10 m physical area = 1 km² canyon-highway zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion
Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System
1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.

Key Locations
Blue HQ Hidden-Gate Commandi — (0..0). 150×150 m Armored vehicles. drone pads. convoy staline.
Red HQ Hidden-Tee Command Soint — (1000.1000). 150×150 m Elevated bunker with radar mast. AA emplacements. and lone-ranae sensors.
Central Canyon Highway — (500.450). 300×300 m High-speed traversal zone: lone sightlines. ambush corners. dust plumes.
Checkpoint Oeost — (250.800). 120×120 m Barricades. fuel tanks. reaair bays. interior flankine routes.
Ridge Network — (800.300. 200×200 m High-ground overwatch. cruise nests. drone launch pads.
Ambush Corridor — (0.450 → 400.900) Primary chokepoint: narrow. exposed. Ideal for ambushes.
Dry Riverbed Route — (100.0 → 450.350) Safer but slow: unstable terrain and limited cover.
Ridge Relav A — (650.1501. +240 m Lona-ranae sensor vantage point.
Ridge Relav B — (150.650). +260 m Comms relay hub and overwatch control node.
Neutral Forward Outoost — (500.950) Drone resupply zone.

Chokepoints & Sightlines

Ambush Corridor
Major chokepoint; controlling it determines convoy movement.

Ridge Network
Extreme long-range sightlines; ideal for overwatch and artillery.

Canyon Highway
High-speed engagements; dust plumes affect accuracy.

Dry Riverbed Route
Predictable but safe; vulnerable to ridge overwatch.

Sightlines
Ridge Relay. A provides long-range sensor coverage (~350 m). Dust storms and heat shimmer create fog-of-war pockets.

Deployment Zones

Blue Deployment SW highway gate + convoy staline
Red Deployment NE cliffs (no mounted + AA emplacements)
Neutral Forward Zone Outpost at (500.950) for drone drops

Assets & Environment

Terrain
Canyons, ridges, highways, dry riverbeds, checkpoints.

Vegetation
Sparse shrubs, desert grass, small tree.

Buildings & Props
Military: AA aunc. radar dishes. bunkers. industrial: Checkpoints. aenerators. oipelines. Logistics: Trucks. crates. fuel drums. drones.

Effects
Dust storms, heat shimmer, mirage distortion, wind gusts.

Audio
Wind, engines, distant artillery, radio chatter.

LOD & Budget

Asset Cate	Count	Tri/Item	Total Tris
Canvon/R	3	200k	600k
Industrial	4	80k	320k
Buildines/	15	50k	750k
Roads/Pa	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget
2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins
Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering
Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors
Single spatial anchor for entire map.

Interaction
Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer
Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones
1. Prototvpe Layout — Month 1 Block-out canyon highway. ridaes. checkpoints. AR anchor test.
2. Core Svstems — Month 2 NavMesh. mobility autonomy behaviors. ambush logic.
3. AR Interaction — Month 3 Anchors. plane detection. basic interactivity.
4. Content Fill — Month 4–5 Final models. textures. liahtina. audio.
5. Gameplav & Polish — Month 6–7 Objectives. balance. VFX (dust storms. heat shimmer).
6. Testina & QA — Month 8 Multiplaver stress test. sensor validation.

Testing Plan

Autonomy
Verify mobility routing, ambush-aware navigation, hazard avoidance.

Comms/Sensors
Test relay destruction, fog-of-war updates, LOS checks under dust and heat distortion.

Performance
≥60 Hz AR render on target device.

QA Checklist
Navigation, mobility logic, deployment budget enforcement, anchor drift, sensor accuracy.